MODULE 3: PROGRAMMING FOR BIOLOGICAL DATA

Session 6: ANOVA and Linear Regression

Comparing Multiple Groups & Building Calibration Curves

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Learning Objectives

ANOVA

Compare means across multiple groups using Analysis of Variance.

Post-hoc

Identify exactly which groups differ using Tukey HSD.

Regression

Build linear regression models for standard calibration curves.

Validation

Assess model fit using R^2 and rigorous residual analysis.

Clinical Context

Method Applications

Method	Application
ANOVA	Compare 3+ analyzers, reagent lots, or operators.
Regression	Build calibration curves and verify assay linearity.

Today's Scenarios

- **Scenario 1:** Are our three chemistry analyzers giving consistent results?
- **Scenario 2:** Is our assay linear across the reportable range according to CLSI EP06?

Part 1: Analysis of Variance (ANOVA)

Comparing Means Across Multiple Groups

When to Use ANOVA

The Right Test for the Job

Scenario	Statistical Test
Compare **2** groups	T-test
Compare **3+** groups	ANOVA

⚠️ Warning

Running multiple t-tests inflates the false positive rate!

- Comparing 3 groups requires 3 separate t-tests.
- This leads to a **~14% chance** of finding a false difference.
- ANOVA controls this error by testing all at once.

Scenario: Analyzer Comparison

The Setup

Your lab has **three chemistry analyzers**: A, B, and C.

You run the same glucose QC material 10 times on each instrument.

Question: Are all three analyzers calibrated correctly?

Preliminary Data

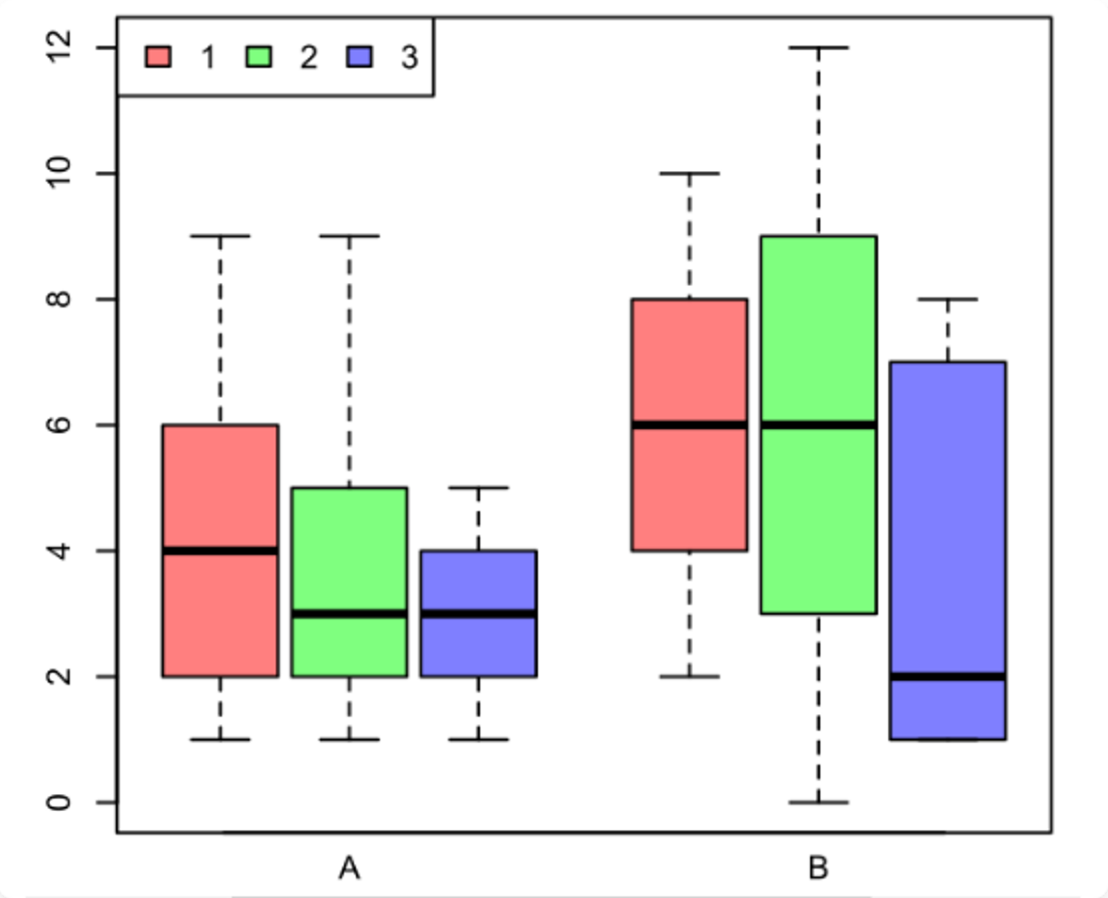
Analyzer	Mean Glucose	SD
A	102.5	1.5
B	98.2	1.4
C	106.8	1.6

Visualizing with Boxplots

```
boxplot(Glucose ~ Analyzer,
        data = analyzer_data,
        col = c("coral",
                "steelblue",
                "seagreen"))

# Add target line
abline(h = 100, col = "red", lty = 2)
```

Visual inspection: The boxplot clearly suggests that the analyzers are producing different results, with C being notably higher.



How ANOVA Works

The F-Statistic

ANOVA compares **between-group variance** (differences between analyzers) to **within-group variance** (precision error).

$$F = \frac{\text{Variance Between Groups}}{\text{Variance Within Groups}}$$

Large F = Groups are different
Small F = Groups are similar

ANOVA Hypotheses

Null Hypothesis (H_0)

All group means are equal.

$$\mu_A = \mu_B = \mu_C$$

Alternative Hypothesis (H_1)

At least one mean differs from the others.

Decision Rule: If $p < 0.05$, we reject H_0.

ANOVA in R

R Code

```
# Fit ANOVA model
anova_result <- aov(Glucose ~ Analyzer,
                    data = analyzer_data)

# View results
summary(anova_result)
```

Output

```
            Df  Sum Sq  Mean Sq  F value    Pr(>F)
Analyzer     2    592      296     120.3    <2e-16
***
Residuals   27     66      2.5
```

Interpreting the ANOVA Table

Column	Meaning
Df	Degrees of freedom (Groups - 1)
Sum Sq	Sum of squares (total variance explained)
Mean Sq	Mean square = Sum Sq / Df
F value	Test statistic (Variance Ratio)
Pr(>F)	**P-value** (The probability of this result if H_0 is true)

$p < 0.05$ → At least one analyzer differs significantly.

The Problem

"ANOVA tells us that 'something' is different, but it doesn't tell us 'what' is different."

*To find the specific culprit, we need a **post-hoc test**.*

Tukey's Honest Significant Difference

Purpose

Perform pairwise comparisons after a significant ANOVA to identify exactly which groups differ.

It controls for the increased error rate of multiple testing.

R Code

```
# Tukey HSD test
TukeyHSD(anova_result)
```

This compares:

- A vs B
- A vs C
- B vs C

Reading Tukey Output

```
              diff    lwr     upr    p adj
B-A          -4.30  -5.89   -2.71   0.0000
C-A           4.30   2.71    5.89   0.0000
C-B           8.60   7.01   10.19   0.0000
```

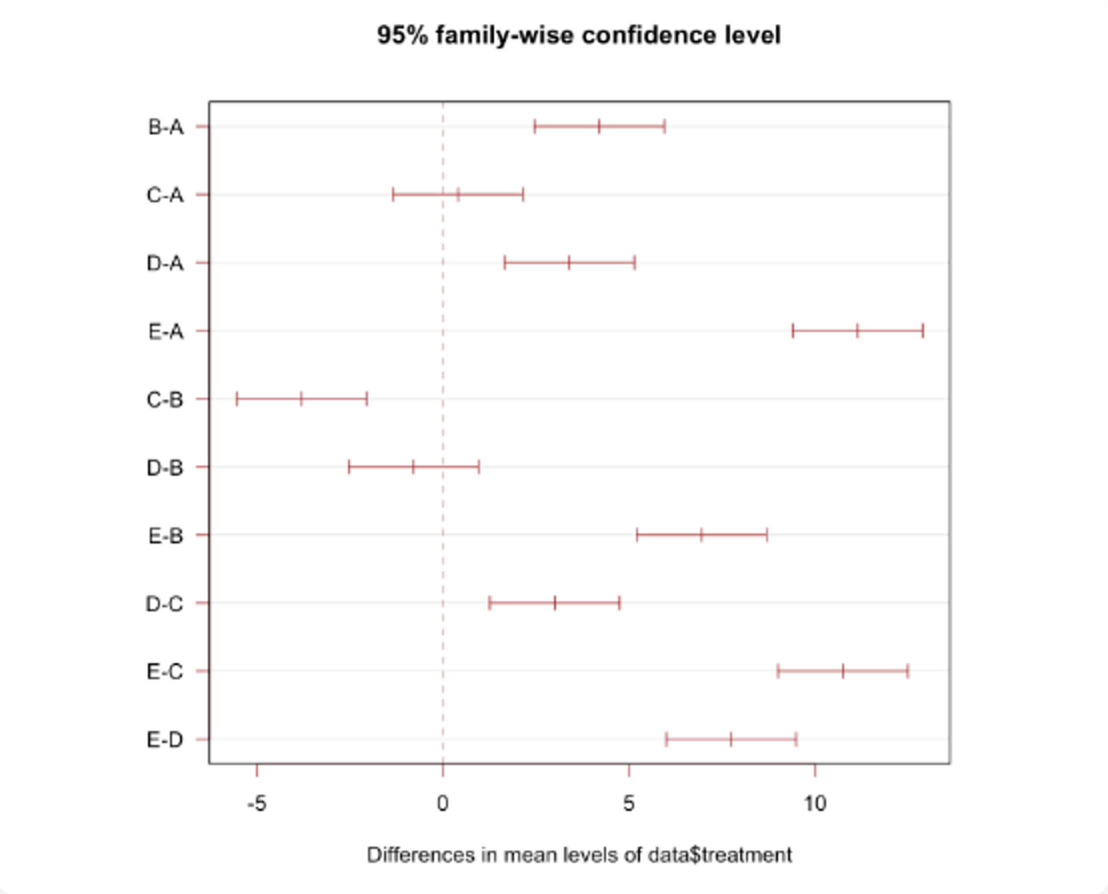
Column	Meaning
diff	Difference between group means.
lwr, upr	95% Confidence Interval of the difference.
p adj	Adjusted p-value.

Tukey Plot

```r
plot(TukeyHSD(anova_result))
abline(v = 0, col = "red", lty = 2)
```

Interpretation:

- The lines represent Confidence Intervals.
- If an interval **crosses zero**, the difference is NOT significant.
- Here, all intervals are far from zero, indicating significant differences for all pairs.



Clinical Action Items

Based on the statistical findings, we take action:

Comparison	Difference	Significant?	Action
B vs A	-4.3	Yes ⚠️	Recalibrate B (reading low)
C vs A	+4.3	Yes ⚠️	Recalibrate C (reading high)
C vs B	+8.6	Yes ⚠️	Major discrepancy detected

Checking ANOVA Assumptions

1

Normality

Residuals should be normally distributed.

```
shapiro.test(residuals(model))
```

2

Homogeneity

Variances should be equal across groups.

```
bartlett.test(Glucose ~ Analyzer)
```

3

Independence

Observations must be independent of each other.

Part 2: Linear Regression & Calibration

Building the Standard Curve

The Foundation of Quantitative Testing

The Concept

Every quantitative assay uses a **calibration curve** to convert measured signals into concentrations.

We model this relationship using the linear equation.

$$y = mx + c$$

Signal = Slope × Concentration + Intercept

CLSI EP06: Linearity Verification

Guideline Requirements

- Test samples across the full **reportable range**.
- Verify the **linear relationship** between signal and concentration.
- Document **recovery** at each level.
- Identify any **saturation or non-linearity**.

Standard for Lab Compliance

Linearity Verification Data

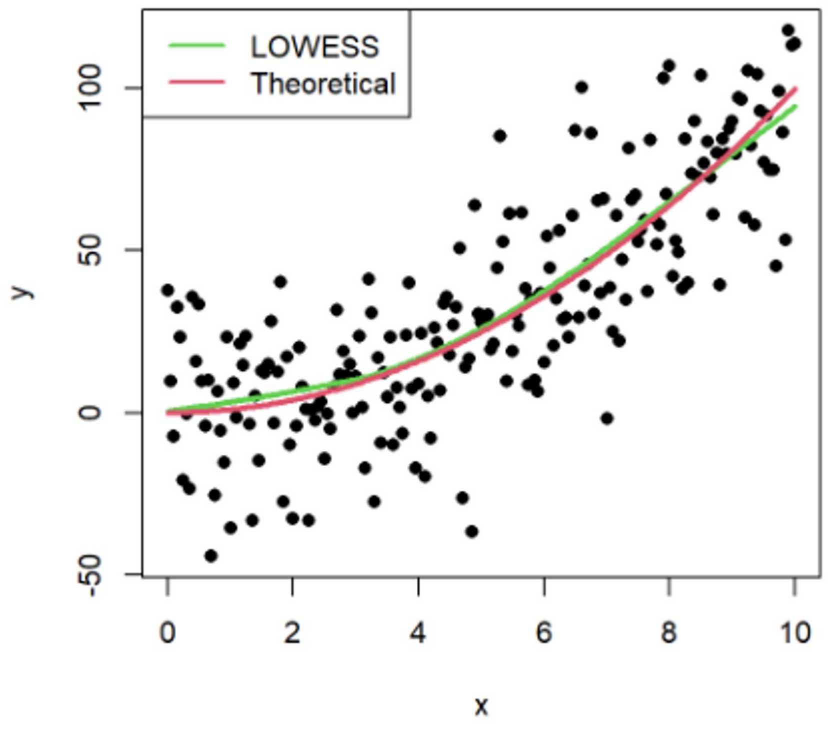
Concentration (mg/dL)	Signal (OD)	Replicates
0 (blank)	0.052	3
25	0.245	3
50	0.478	3
100	0.952	3
200	1.875	3
400	3.425	3

Visualizing Calibration Data

```
plot(Concentration, Signal,
     main = "Linearity Verification",
     pch = 19,
     col = "steelblue")
```

Before fitting a model, always perform a visual check.

Does the relationship look linear?



Linear Regression in R

Fitting the Model

We use the lm() function (linear model).

```
model <- lm(Signal ~ Concentration,
            data = linearity_data)
```

Viewing Results

The summary() function provides coefficients and fit statistics.

```
summary(model)
```

Interpreting lm() Output

```
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)   0.05200    0.01500    3.47  0.00312 **
Concentration 0.00872    0.00008  109.00   <2e-16 ***

Multiple R-squared: 0.9987
```

Term	Value	Meaning
Intercept	0.052	The background signal when concentration is 0.
Slope	0.00872	Signal increase per unit of concentration.
R^2	0.9987	99.87% of the variance is explained by the model.

Getting the Equation

R Code

```
intercept ← coef(model)[1]
# 0.052

slope ← coef(model)[2]
# 0.00872

r_sq ← summary(model)$r.squared
# 0.9987
```

Final Calibration Equation:

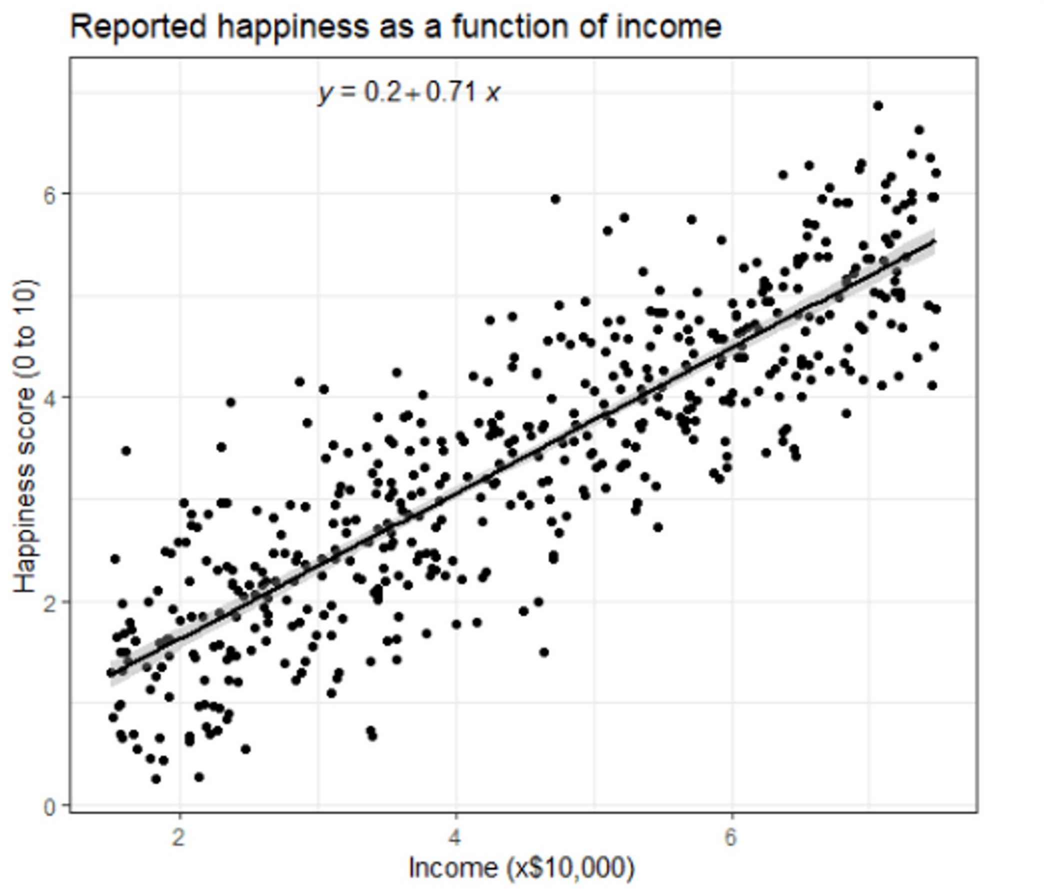
$$\text{Signal} = 0.00872 \times \text{Concentration} + 0.052$$

Adding the Regression Line

```
plot(Concentration, Signal,
     pch = 19, col = "steelblue")

# Add the regression line
abline(model, col = "red", lwd = 2)
```

This visualizes how well our calculated model fits the actual data points.



Understanding R^2

Coefficient of Determination

$$R^2 = 1 - \frac{SS_{residual}}{SS_{total}}$$

Interpretation

R^2 Value	Meaning
0.99+	Excellent fit (Standard for Clinical Labs)
0.95–0.99	Good fit
< 0.90	Poor fit — Investigate!

Why Check Residuals?

The Concept

Residual = Observed - Predicted

Residuals allow us to see patterns that R^2 might hide.

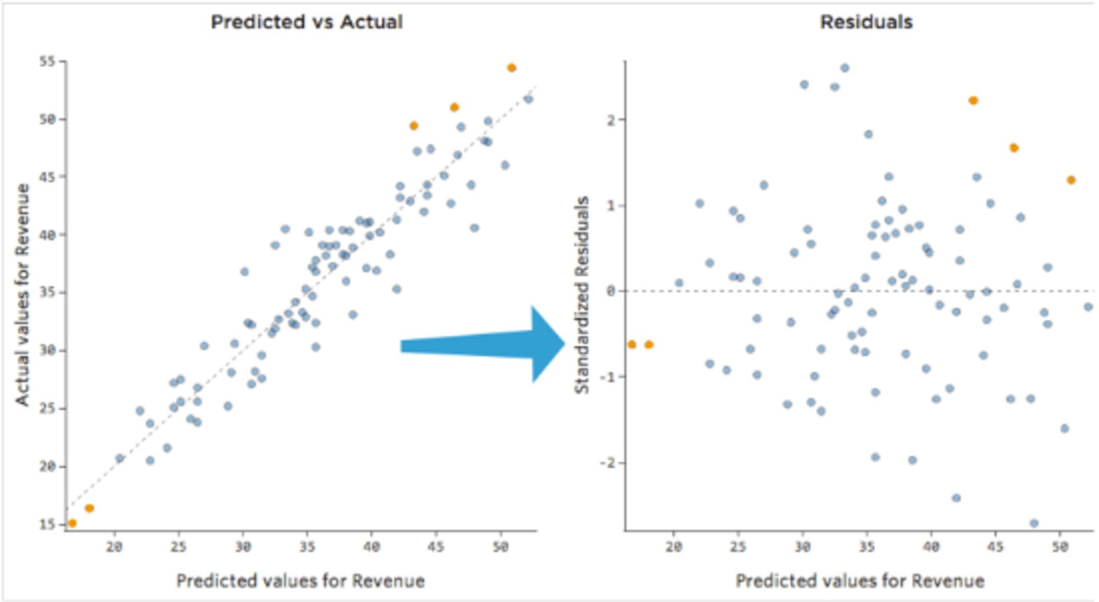
What to Look For

- ✓ **Random Scatter:** Good model.
- ❗ **Curvature:** Non-linearity (needs polynomial fit).
- ❗ **Funnel Shape:** Heteroscedasticity (unequal variance).

Creating a Residual Plot

```
plot(Concentration, residuals(model),
     main = "Residual Plot",
     pch = 19)

# Add zero line
abline(h = 0, col = "red", lty = 2)
```



Note that we've colored in a few dots in orange so you can get the sense of how this transformation works.

Built-in Diagnostic Plots

```
par(mfrow = c(2,2))
plot(model)
```

What R Generates

Plot	Checks For
Residuals vs Fitted	Linearity
Q-Q Plot	Normality of Residuals
Scale-Location	Homoscedasticity
Residuals vs Leverage	Influential Outliers

Using the Calibration Curve

$$\text{Concentration} = \frac{\text{Signal-Intercept}}{\text{Slope}}$$

R Code

```
unknown_signal ← 1.5

predicted_conc ← (unknown_signal -
intercept) / slope

# Result: 166.1 mg/dL
```

Linearity Acceptance Criteria

Criterion	Typical Requirement
R^2	≥ 0.99
Residuals	Within ±10% of predicted value
Recovery	95–105% at each concentration level

Note: Specific requirements vary by analyte and method SOP.

Automated Verification Check

if (r_squared >= 0.99) { cat("✅ R² PASS") } else { cat("⚠️ R² FAIL") } if (max(abs(residuals)) <= 0.10 * predicted) { cat("✅ Residuals PASS") } else { cat("⚠️ Residuals FAIL") }

Summary: ANOVA vs. Regression

Feature	ANOVA	Linear Regression
Purpose	Compare group means	Model relationships
Question	"Are groups different?"	"What is the equation?"
R Function	aov()	lm()
Key Statistics	F-value, p-value	R^2, slope, intercept
Lab Use	Analyzer/Lot Comparison	Calibration Curves

Key Takeaways

ANOVA

Use for comparing 3+ groups. Always follow up with Tukey HSD.

Regression

Models $y = mx + c$. Use $R^2 \geq 0.99$ as a quality standard.

Residuals

Essential for validating that your model assumptions are met.

Tutorial 6

Calibration Verification in Practice

Open your R environment

Tutorial 6: Calibration Verification

Building and Validating Standard Curves

Your Scenario

The Task

You are performing **annual linearity verification** following CLSI EP06 guidelines.

- Verify linear response across reportable range.
- Document calibration equation.
- Ensure $R^2 \geq 0.99$.

The Data

6 Concentration Levels

Each measured in **triplicate**.

What You'll Practice

Step	Skill	Time
1-2	Load and explore data	10 min
3	Fit linear model	15 min
4	Create calibration plot	15 min
5	Analyze residuals	15 min
6-7	Calculate unknowns & Report	15 min

Steps 1–2: Load and Visualize

Load Data

```
linearity_data ← data.frame(
  Concentration = rep(c(0, 25, 50, 100, 200,
400), each = 3),
  Signal = c(0.052, 0.048, ... )
)
```

Visualize

```
str(linearity_data)

aggregate(Signal ~ Conc, FUN = mean)

plot(Concentration, Signal)
```

Step 3: Fit Linear Model

The Magic Line

```
model ← lm(Signal ~ Concentration,
            data = linearity_data)

summary(model)
```

Extract Parameters

```
intercept ← coef(model)[1]
slope ← coef(model)[2]
r_squared ← summary(model)$r.squared
```

Step 4: Calibration Plot

Base R

```
plot(Concentration, Signal, pch = 19)
abline(model, col = "red", lwd = 2)
```

ggplot2 (Publication Quality)

```
ggplot(data, aes(x=Concentration,
y=Signal)) +
  geom_point(size = 3) +
  geom_smooth(method = "lm", se = TRUE)
```

Step 5: Check Residuals

Why?

Residuals reveal problems R^2 misses:

- Curvature = Non-linearity
- Funnel = Unequal variance

Code

```
plot(Concentration, residuals(model))
abline(h = 0, col = "red", lty = 2)
```

Goal: Random scatter around zero.

Step 6: Predict Unknowns

$$\text{Conc} = \frac{\text{Signal-Intercept}}{\text{Slope}}$$

Calculation

```
unknown_signal ← 1.534

predicted_conc ← (unknown_signal -
intercept) / slope
```

Step 7: Verification Report

Checklist

- Date and Analyst
- Calibration Equation
- R^2 Value (Pass/Fail)
- Conclusion

Example Conclusion

```
"Linearity verification PASSED. R² = 0.9987 exceeds acceptance criterion of 0.99. Assay is acceptable for clinical use."
```

Questions?

Session 6 Complete